

Problem Set 8
COMP301 Spring 2025
18.04.2024 - 20.04.2025

Instructions:

- Submit your answers to the Blackboard PS8 assignment until 20.04.2024 Saturday, at 23.59.
- Submit your PDF file to learnhub. *yourIDno_username.zip*. (Example: *123456_aku-tuk21.zip*)
- Read the questions carefully. Good luck!

Problem 1: Determine whether the PROC codes given below run without error or not. If they run, evaluate their results and if they do not, state the reason for the error.

a). let x = 5
 in let y = proc (z) -(t, z)
 in let t = 5
 in (y 3)

b). let x = 1
 in let y = proc (z) -(z, x)
 in let x = 7
 in let z = 5
 in (y x)

c). let x = 9
 in let y = proc (z) (z x)
 in let x = 0
 in let t = proc (x) if zero?(x) then 5 else 3
 in (y t)

Problem 2: In this part, you will draw the state of the environment for a given expression. Also, calculate the output of the code pieces when executed.

(1)

```
1  let z = 2
2      in let f = proc(x) if zero? (x) then 1
3                          else -(x, 3)
4                          in (f 6)
```

Draw the environment state diagram of the given code at the instant where value-of is called for «-(x, 3)» at line 3.



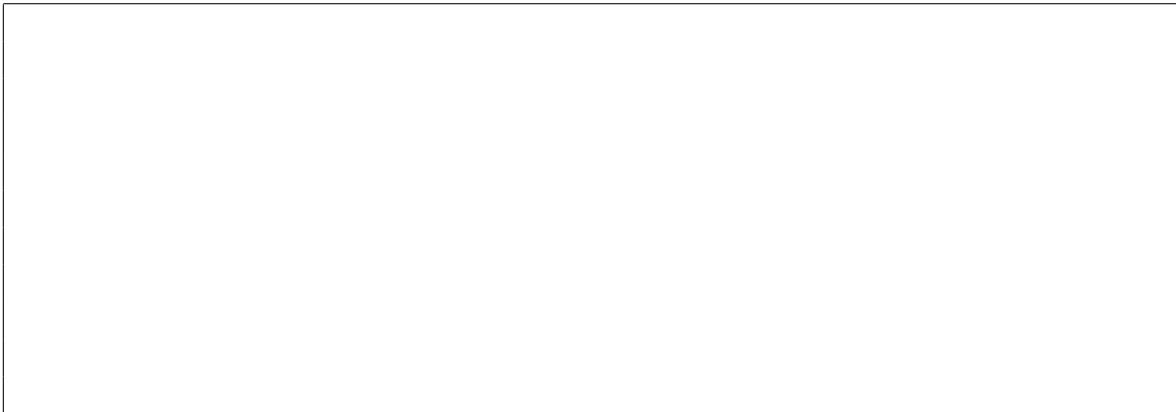
(2)

```

1  let x = 4
2      in let f = proc (y) -(y, x)
3          in let z = 3
4              in (f 6)

```

Draw the environment state diagram of the given code at the instant where `value-of` is called for « `-(y, x)` » at line 2.



(3)

```

1  let a = 3
2      in let p = proc (x) -(x, a)
3          in let a = 5
4              in -(a, (p 2))

```

Draw the environment state diagram of the given code at the instant where `value-of` is called for « `-(a, (p 2))` » at line 4.

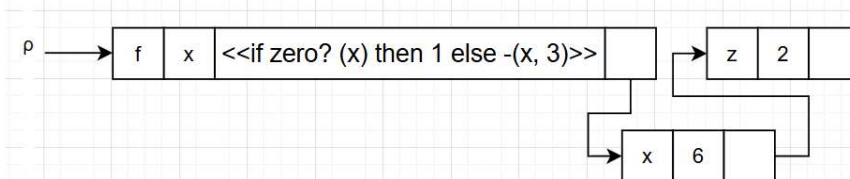


Problem 1

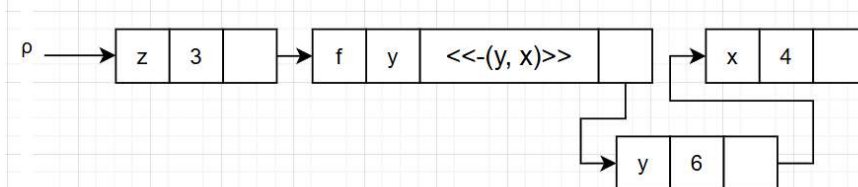
- (a) `let x = 5 in let y = proc (z) -(t, z) in let t = 5 in (y 3)`
 Error: t is undefined. y cannot be defined before t.
- (b) `let x = 1 in let y = proc (z) -(z, x) in let x = 7 in let z = 5 in (y x)`
 6
- (c) `let x = 9 in let y = proc (z) (z x) in let x = 0 in let t = proc (x) if zero?(x) then 5 else 3 in (y t)`
 3

Problem 2

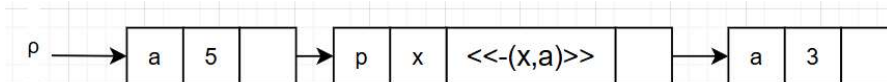
- 1) `let z = 2 in let f = proc(x) if zero? (x) then 1 else -(x, 3) in (f 6)`
 Environment at the instant where value-of is called for «-(x,3)»



- 2) `let x = 4 in let f = proc(y) -(y, x) in let z = 3 in (f 6)`
 environment at the instant where value-of is called for « -(y, x) »



- 3) `let a = 3 in let p = proc (x) -(x,a) in let a = 5 in -(a, (p 2))`
 environment at the instant where value-of is called for «-(a, (p 2))»



Note: when value-of is called for `-(a, (p 2))`, 2 is not added to the env of p yet